



Binomial Distribution

Part A: Binomial Distribution

3unit

selective

Q 1

Blacktown Boys

Bags of lollipops are supposed to contain 50 lollipops. Production records indicate that 92% of bags contain 50 lollipops. A batch of 15 bags is sampled. If more than 2 bags do not contain exactly 50 lollipops, production is stopped.

- Find the probability, correct to four decimal places, that exactly 2 of the 15 bags selected do not contain 50 lollipops. 1
- Find the probability, correct to four decimal places, that production is stopped. 2

$$\begin{aligned} i \quad P(X=2), X \sim B(15, 0.18) \\ {}^{15}C_2 \times 0.08^2 \times 0.92^{13} \\ = 0.227306 \dots \\ = 0.2273 \end{aligned}$$

$$\begin{aligned} ii \quad \text{Production is stopped} = P(X > 2) \\ = 1 - P(X \leq 2) \\ = 1 - (P(X=0) + P(X=1) + P(X=2)) \\ = 1 - ({}^{15}C_0 \times 0.08^0 \times 0.92^{15} + {}^{15}C_1 \times 0.08^1 \times 0.92^{14} + {}^{15}C_2 \\ \times 0.08^2 \times 0.92^{13}) \\ = 0.112965 \dots = 0.1130 \end{aligned}$$

Q 2

Bonnyrigg

Ms. Nguyen gives snacks to her students every now and then. Her class has worked out that the probability that she will do so on any given day is $\frac{2}{3}$. The students record the number of times Ms. Nguyen gives snacks over a 10-day period. What is the probability that she will give snacks on 8 or more of these 10 days?

- A. 0.1950 B. 0.2990 **C. 0.2991** D. 0.1951

Q 3

Bonnyrigg

An event has a probability of success of $\frac{1}{3}$ in a single trial. If n trials are conducted, find the least value of n for which the probability of obtaining exactly four successes exceeds triple the probability of obtaining exactly three successes. 3

$$\begin{aligned} P(r=3) &= {}^nC_3 \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^{n-3} \\ P(r=4) &= {}^nC_4 \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^{n-4} \\ {}^nC_4 \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^{n-4} &> {}^nC_3 \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^{n-3} \times 3 \end{aligned}$$

$$\begin{aligned} \frac{n!}{4!(n-4)!} \times \frac{1}{3} &> 3 \times \frac{n!}{3!(n-3)!} \times \frac{2}{3} \\ \frac{(n-3)!}{(n-4)!} &> 24 \quad \checkmark \textcircled{1} \\ n-3 &> 24 \\ n &> 27 \\ \text{Smallest value } n &= 28 \end{aligned}$$

Q 4

Bonnyrigg

An archer can successfully hit a target with a probability of 0.9. The archer attempts to hit the target 80 times. The outcome of each attempt is independent of any other attempt. Given that the archer successfully hits the target at least 70 times, the probability that the archer successfully hits the target exactly 74 times, correct to four decimal places, is

- A. 0.3635 B. 0.8266 **C. 0.1494** D. 0.3005 E. 0.1701

See that $np > 5$ and $nq > 5$:

$$X \sim N(72, 7.2), \mu = 72, \sigma = 2.683$$

$$\begin{aligned} P(X \geq 70) &= P(X \geq 69.5) = P\left(Z \geq \frac{69.5 - 72}{2.68}\right) \\ &= P(Z \geq -0.933) = P(Z \leq 0.933) \approx 0.8264 \end{aligned}$$

(used $z = 0.94$ on the table)

$$X \sim \text{Bin}(80, 0.9)$$

$$\begin{aligned} P(X=74) &= {}^{80}C_{74} \times 0.9^{74} \times 0.1^6 = 0.1235 \\ \therefore \frac{0.1235}{0.8264} &= 0.1494 \rightarrow C \end{aligned}$$

Q 5

CSSA

A luxury bed and breakfast has 6 standard and 5 executive rooms. On any day the probability that a standard room is unoccupied is 6.4% and the probability that an executive room is unoccupied is 13.1%. On any random day, find the probability that 2 standard rooms and 2 executive rooms are unoccupied. 3

$$\text{Standard Room: } X \sim B(6, 0.064)$$

$$\begin{aligned} P(X=2) &= {}^6C_2 (0.064)^2 (0.936)^4 \\ &= 0.0471 \quad (4 \text{ dp}) \end{aligned}$$

$$\text{Executive Room: } X \sim B(5, 0.131)$$

$$\begin{aligned} P(X=2) &= {}^5C_2 (0.131)^2 (0.869)^3 \\ &= 0.1126 \quad (4 \text{ dp}) \end{aligned}$$

$$P(2 \text{ Standard and } 2 \text{ Executive unoccupied}) = 0.0471 \times 0.1126 = 0.0053 \quad (4 \text{ dp})$$

Q 6

CSSA

There are eight questions in a multiple-choice test. Each question has four possible answers, only one of which is correct.

What is the probability of answering exactly five questions correctly by chance alone, correct to 3 significant figures?

- A. 0.0000153 B. 0.000412 **C. 0.000977** **D. 0.0231**

$$P(\text{exactly } 5 \text{ correct}) = {}^8C_5 \left(\frac{1}{4}\right)^5 \left(\frac{3}{4}\right)^3 = \frac{189}{8192} = 0.02307128906 \approx 0.0231$$



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Q 7

Enterprises

It is known 5% of light bulbs produced by a local company are faulty. What is the probability that in a random sample of 7 light bulbs, exactly three light bulbs are faulty.

$$P(\text{exactly three light bulbs are faulty}) \\ = {}^7C_3 (0.95)^4 (0.05)^3 = 0.00356346$$

Q 8

Hurlstone

The probability that the 7:30 a.m. train arrives on time is 0.85.

Find an expression for the probability that the train is on time at least five days during one week

Let p be the probability of the train being on time
 $p = 0.85, n = 7 \quad P(X = x) = {}^7C_x 0.85^x 0.15^{7-x}$

Expression is:

$$P(X \geq 5) = P(X = 5) + P(X = 6) + P(X = 7) \\ = {}^7C_5 0.85^5 0.15^2 + {}^7C_6 0.85^6 0.15^1 + {}^7C_7 0.85^7 0.15^0$$

Q 9

Manly

Which one of the following describes a Bernoulli random variable?

- A. The times achieved by athletes in a 100-metre race
- B. The value of a card chosen from a pack of 52 playing cards
- C. Choosing a red pen from a container that contains red, black and blue pens
- D. The number of emails a doctor receives in a week

Limited to two options

Red or Not Red

Q10

East Hills Boys

2 A game is played in which players throw netballs, aiming to get them through the hoop. The players are given two options in how they play the game.

Option 1: Players take 2 attempts and win a prize if at least one of the balls goes through.

Option 2: Players take 3 attempts and win a prize if at least two of the balls go through.

Karen decides to play.

The probability that she will get a ball through on any attempt is p , where $p > 0$.

- 2 i) Show that the probability that Karen wins if she takes Option 1 is $2p - p^2$ 1
- ii) Show that the probability that Karen wins if she takes Option 2 is $3p^2 - 2p^3$. 1
- iii) Karen has calculated that she is 3 times as likely to win by choosing Option 1 over Option 2. Find the value of p . 2

(i) Probability of at least 1 netball going in includes 1 going in or 2 going in.

$$P(\text{at least 1}) = P(1) + P(2) \\ = {}^2C_1 p(1-p) + p^2 \\ = 2p - 2p^2 + p^2 \\ = 2p - p^2$$

OR Probability of at least 1 netball going in is 1 minus the probability of none going in

$$P(\text{at least 1}) = 1 - P(0) = 1 - (1-p)^2 = 1 - (1 - 2p + p^2) = 2p - p^2$$

ii) Probability of at least 2 netballs going in includes 2 going in or 3 going in.

$$P(\text{at least 2}) = P(2) + P(3) = {}^3C_2 p^2(1-p) + p^3 = 3p^2 - 3p^3 + p^3 = 3p^2 - 2p^3$$

iii) Let $2p - p^2 = 3(3p^2 - 2p^3)$

$$2p - p^2 = 9p^2 - 6p^3 \\ 6p^3 - 10p^2 + 2p = 0 \\ 2p(3p^2 - 5p + 1) = 0$$

$$2p = 0 \quad \text{or} \quad 3p^2 - 5p + 1 = 0$$

$$p = 0 \quad \text{or} \quad p = \frac{5 \pm \sqrt{5^2 - 4(3)(1)}}{6} = 1.434258546 \quad \text{or} \quad 0.2324081208$$

$$\text{but } 0 < p < 1 \therefore p = 0.2324081208 \approx 0.23$$

$$\text{Or } p = \frac{5 + \sqrt{13}}{6} \text{ since } 0 < p < 1$$



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Q11

Penrith

Which of the following statement does *not* describe a Bernoulli random variable?

- A. Selecting a faulty phone in a random quality control check.
- B. Guessing the correct answer to a true/false question.
- C. The number of tails when tossing 2 coins.**
- D. A missile hitting a target.

Q12

Sydney Tech

A coin is biased such that the probability of a head is 0.8. The probability that exactly four tails will be observed when the coin is flipped ten times is:

- (A) $10 \times 0.2^6 \times 0.8^4$
- (B) $^{10}C_4 \times 0.2^4 \times 0.8^6$**
- (C) $^{10}C_4 \times 0.2^6 \times 0.8^4$
- (D) $10 \times 0.2^4 \times 0.8^6$

Q13

Sydney Tech

A standard die is rolled 5 times. What is the probability of rolling a four at least two times? 2

$$P(\text{At least two}) = 1 - \left({}^5C_0 \left(\frac{5}{6}\right)^5 + {}^5C_1 \left(\frac{1}{6}\right) \times \left(\frac{5}{6}\right)^4 \right)$$

$$= \frac{763}{3888}$$

Q14

Newington

A multiple-choice test contains ten questions. Each question has four choices for the correct answer. Only one of the choices is correct.

- (i) What is the probability of getting 70% correct with random guessing? 1
- (ii) What is the probability of getting at most 70% correct with random guessing? 2

(i) Let p be the probability of getting the correct answer.

$$p = \frac{1}{4}, n = 10$$

$$P(X = x) = {}^{10}C_x \left(\frac{1}{4}\right)^x \left(\frac{3}{4}\right)^{10-x}$$

$$P(X = 7) = {}^{10}C_7 \left(\frac{1}{4}\right)^7 \left(\frac{3}{4}\right)^{10-7} = \frac{405}{131072}$$

$$(ii) P(X \leq 7) = 1 - (P(8) + P(9) + P(10))$$

$$= 1 - \left({}^{10}C_8 \left(\frac{1}{4}\right)^8 \left(\frac{3}{4}\right)^{10-8} + {}^{10}C_9 \left(\frac{1}{4}\right)^9 \left(\frac{3}{4}\right)^{10-9} + {}^{10}C_{10} \left(\frac{1}{4}\right)^{10} \left(\frac{3}{4}\right)^{10-10} \right)$$

$$= 0.99958...$$

Q15

Manly

On average an archer scores a bullseye once in every three shots at the target. A competitor is allowed eight shots at the target.

- i) Find the probability that a bullseye is hit exactly twice. 1
- ii) Find the probability that a bullseye is hit at least twice. 1
- iii) How many shots should the archer use so as to be 90% certain that at least one bullseye is hit? 2

$$p_{\text{bullseye}} = \frac{1}{3} \quad p = \frac{1}{3} \quad q = \frac{2}{3}$$

$$p_2 = {}^8C_2 p^2 q^6 = \frac{1792}{6561}$$

ii At least 2 = 1 - (0 or 1 hits)

$$1 - {}^8C_0 \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^8 - {}^8C_1 \left(\frac{1}{3}\right)^1 \left(\frac{2}{3}\right)^7 = \frac{5281}{6561}$$

iii no success < 0.1

$$\left(\frac{2}{3}\right)^n < 0.1$$

$$n \log_e \left(\frac{2}{3}\right) < \log_e (0.1)$$

$$n > \frac{\log(0.1)}{\log\left(\frac{2}{3}\right)}$$

$$n > 5.67$$

$$n = 6$$

Q16

Total Education

Two basketball teams, the Lions and the Tigers, take part in the play-off for the championship. The team which first wins three games will be the champion.

From past encounters of the two teams, it is estimated the Lions has a probability of 0.6 of beating the Tigers and a probability of 0.4 of losing, assuming there are no draws.

- (i) Explain why the maximum number of games to be played is five. 1
- (ii) Find the probability of determining the championship in only three games. 1
- (iii) Find the probability of determining the championship in exactly four games. 3

(i) If the championship is not determined after 4 games, then each team has not won 3 games. Thus each team has won 2 games each and so the 5th game will be the decider.

$$(ii) \text{ In 3 games } P(3) = (0.6)^3 + (0.4)^3$$

$$= 0.28$$

(iii) For the Lions to win 3 games and lose 1 in 4
The probability is ${}^4C_3 (0.6)^3 (0.4)$
but this includes the case of winning the first 3 games and not the 4th which must be excluded so $P(\text{4 game championship})$
[i.e. 1st 3 games is a win] $= {}^4C_3 (0.6)^3 (0.4) - (0.6)^3 (0.4)$
 $= 0.2592$
Sim for Tigers $= {}^4C_3 (0.4)^3 (0.6) - (0.4)^3 (0.6)$
 $= 0.1152$
Required prob $= 0.2592 + 0.1152 = 0.3744$





Q 1

Bonnyrigg

A recent census showed that 20% of the adults in a city eat out regularly.

- (i) A survey of 100 adults in this city is to be conducted to find the proportion who eat out regularly. Show that the mean and standard deviation for the distribution of sample proportions of such surveys are 0.2 and 0.04 respectively. 2
- (ii) Use the extract shown from a table giving values of $P(Z < z)$, where z has a standard normal distribution, to estimate the probability that a survey of 100 adults will find that at most 15 of those surveyed eat out regularly. 2

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177

i. $\bar{x} = np = 20$, $\sigma^2 = np(1-p) = 16$ so $\sigma = 4$

$\therefore \bar{x}_{\text{proportion}} = \frac{20}{100} = 0.2$ $\sigma_{\text{proportion}} = \frac{4}{100} = 0.04$

ii. $P(Z < (0.15 - 0.20) / 0.04) = P(Z < -1.25)$
 $= 1 - P(z < 1.25)$ so estimate is
 $1 - 0.8944 = 0.1056$.

Q 2

Bonnyrigg

The weights of packets of lollies are normally distributed with a mean of 200 g.

If 97% of these packets of lollies have a weight of more than 190 g, then the standard deviation of the distribution correct to one decimal place is

- A. 3.3 g **B. 5.3 g** C. 6.1 g D. 9.4 g E. 12.1 g

$$P\left(Z > \frac{190 - 200}{\sigma}\right) = 0.97$$

$$\rightarrow -\frac{10}{\sigma} = -1.88$$

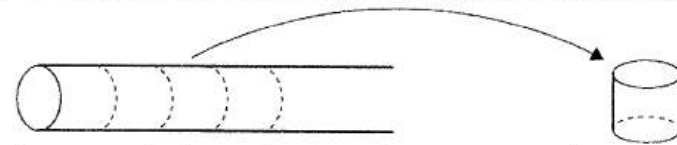
i.e. $\sigma = 5.3$, $\therefore B$

Q 3

Bonnyrigg

A machine produces chocolate in the form of a continuous cylinder of radius 0.5 cm. Smaller cylindrical pieces are cut parallel to its end, as shown in the diagram below.

The lengths of the pieces vary with a mean of 3 cm and a standard deviation of 0.1 cm.



- a. Find the expected volume of a piece of chocolate in cm^3 . 1
 b. Find the variance of the volume of a piece of chocolate in cm^6 . 1
 c. Find the expected surface area of a piece of chocolate in cm^2 . 1

(a) Let $V :=$ volume of cylinder

$$E(V) = \pi \times 0.5^2 \times 3 = \frac{3}{4} \pi u^3$$

$$\begin{aligned} \text{Var}(\pi r^2 h) &= (\pi r^2)^2 \text{Var}(h) = \frac{\pi^2}{16} \times \frac{1}{100} \\ &= \frac{\pi^2}{1600} \end{aligned}$$

(b) Note that the following theory is not mentioned in the syllabus but has appeared in textbooks: $\text{Var}(aX + b) = a^2 \text{Var}(X)$

(c) $E(SA) = 2\pi(0.5)^2 + 2\pi(0.5)(3) = \frac{\pi}{2} + 3\pi u^2$

Q 4

Caringbah

The discrete random variable X has a binomial distribution with n independent trials with probability of success p . If its mean is 6 and its standard deviation is 2, find the values of n and p . 3

$$\mu = 6 \Rightarrow np = 6 \quad (1)$$

$$\sigma^2 = 4 \Rightarrow npq = 4 \quad (2)$$

$$(2) \div (1)$$

$$q = \frac{2}{3}$$

$$\therefore p = \frac{1}{3} \text{ and } n = 18$$





Q 5

Caringbah

- (i) Records show that 64% of students at a school travelled to and from school by bus. Samples of 100 students at the school are taken to determine the proportion who travel to and from school by bus. Show that the distribution of such sample proportions has mean 0.64 and standard deviation 0.048.
- (ii) Use the table (shown below) of $P(Z < z)$, where Z has a standard normal distribution, to estimate the probability that a sample of 100 students will contain at least 58 and at most 64 students who travel to and from school by bus.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177

(b) For students in the sample, the number of students travelling to and from school by bus is a random variable X with the Binomial distribution $B(100, 0.64)$. Hence the sample proportion $\frac{X}{100}$ has mean $\mu = 0.64$ (since $E(\frac{X}{n}) = \frac{np}{n} = p$)

and standard deviation $\sigma = \sqrt{0.64 \times (1-0.64)} = 0.048$

(since $VAR(\frac{X}{n}) = \frac{npq}{n^2} = \frac{pq}{n}$)

(iii) $P(0.58 \leq \frac{X}{100} \leq 0.64)$

Z scores for 0.64 = $\frac{0.64 - 0.64}{0.048} = 0$

Z scores for 0.58 = $\frac{0.58 - 0.64}{0.048} = -1.25$

$\approx P(-1.25 \leq Z \leq 0)$

= $P(Z \leq 1.25) - 0.5 = 0.8944 - 0.5 = 0.3944$

Q 6

East Hills Boys

- Chris is planning to survey 36 people. He will capture the birth month of each person in the data.
- i) What number of people born in January would Chris expect in this sample? (Assume each month has an equal probability of being selected.) 1
- ii) Calculate the probability that the number of people in Chris' sample born in January is within one standard deviation of the mean. 2

$$i) E(X) = np = 36 \times \frac{1}{12} = 3$$

So we would expect 3 people born in January in a sample of 36.

$$ii) \sigma = \sqrt{npq} = \sqrt{36 \times \frac{1}{12} \times \frac{11}{12}} = 1.658312395 \approx 1.66$$

$$3 - \sigma = 1.341687605$$

$$3 + \sigma = 4.658312395$$

So we are looking for the probability that we will have from 2 to 4 people born in January in Chris' sample.

$$P(2 \leq X \leq 4) = P(2) + P(3) + P(4)$$

$$= {}^{36}C_2 \left(\frac{1}{12}\right)^2 \left(\frac{11}{12}\right)^{34} + {}^{36}C_3 \left(\frac{1}{12}\right)^3 \left(\frac{11}{12}\right)^{33} + {}^{36}C_4 \left(\frac{1}{12}\right)^4 \left(\frac{11}{12}\right)^{32} = 0.6364946828$$

Q 7

Gosford

An examination consists of 30 multiple-choice questions, each question having five possible answers. A student guesses the answer to every question. Let X be the number of correct answers. What is $E(X)$?

- (A) 5 (B) 6 (C) 9 (D) 15



Q 8

Gosford

On a roulette wheel, there are 18 red numbers, 18 black numbers, and 1 green number. A ball is dropped onto the spinning wheel and lands on one of the numbers randomly. Each result is independent. A gambler bets that the ball will land on any of the black numbers.

- Define the gambler's bet as a Bernoulli random variable X and give its mean and variance. 1
- If the gambler makes the same bet five times, let the random variable Y be the number of times the gambler wins. Describe the distribution of Y , and give its mean and variance. 2
- If the gambler makes the same bet five times, what is the probability he will win more times than he loses? Give your answer correct to three decimal places. 2

(i) This is a Bernoulli random variable with $p = 18/37$. This may be written

$$X \sim \text{Ber}\left(\frac{18}{37}\right) \text{ or } X \sim \text{Bin}\left(\frac{18}{37}, 1\right)$$

$$\mu = \frac{18}{37}, \sigma^2 = \frac{18}{37} \times \frac{19}{37} = \frac{342}{1369}$$

(ii) This is a binomial random variable with $p = 18/37$ and $n = 5$. This may be written

$$Y \sim \text{Bin}\left(\frac{18}{37}, 5\right)$$

$$\mu = 5 \times \frac{18}{37} = \frac{90}{37} \approx 2.43$$

$$\sigma^2 = 5 \times \frac{18}{37} \times \frac{19}{37} = \frac{1710}{1369} \approx 1.25$$

Q 9

Independent

The discrete random variable X has a Binomial distribution with parameters n and p . If its mean is 6 and its standard deviation is 2, find the values of n and p .

$$\mu = 6 \Rightarrow np = 6 \quad (1)$$

$$\sigma^2 = 4 \Rightarrow npq = 4 \quad (2)$$

$$(2) \div (1) \Rightarrow q = \frac{2}{3}$$

$$\therefore p = \frac{1}{3}, n = 18$$

Q10

James Ruse

The probability of success in a Bernoulli trial is 0.3. What is the variance?

- (A) 0.09 (B) 0.7 (C) 0.3 (D) 0.21

Q11

James Ruse

A random variable X is defined by $P(X = k) = \binom{15}{k} (0.29)^k (0.71)^{15-k}$ for $k = 0, 1, 2, \dots, 15$.

What is the mean of X ?

- (A) 0.29 (B) 0.71 (C) 4.35 (D) 10.65

Q12

Killara

An examination consists of 30 multiple-choice questions, each question having five possible answers. A student guesses the answer to every question. Let X be the number of correct answers. What is $E(X)$?

- (A) 5 (B) 6 (C) 9 (D) 15

$$p = \frac{1}{5} \quad q = \frac{4}{5}$$

$$E(X) = 30 \times \frac{1}{5} = 6$$

Q13

Newington

A binomial random variable X has a mean of 15 and a variance of 10. What are the parameters n and p ?

$$E(X) = np = 15 \quad (1)$$

$$\text{Var}(X) = np(1-p) = 10 \quad (2)$$

Substituting equation (1) into (2)

$$15 \times (1-p) = 10$$

$$1-p = \frac{10}{15} = \frac{2}{3} \text{ or } p = \frac{1}{3}$$

Substituting $p = \frac{1}{3}$ into equation (1)

$$n \times \frac{1}{3} = 15$$

$$n = 45$$

$\therefore$ Parameters are $n = 45$ and $p = \frac{1}{3}$



Q14

Manly

The manufacturers of a chocolate chip cookie are concerned about falling sales. They believe that a fault in their machine means that 15% of the cookies do not have the required number of chocolate chips in them.

A sample of 10 boxes, each containing 20 cookies, is selected.

- Find the mean and standard deviation for unsatisfactory cookies from the sampling distribution.
(State your answer for the standard deviation to four decimal places.)
- What is the probability that more than 20% of the cookies in the sample will be unsatisfactory? (You may assume that the sample of cookies is approximately normally distributed and use a table of z-scores.)

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817

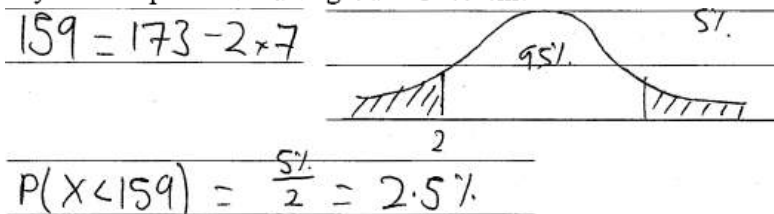
$$\begin{aligned}
 \text{i } \mu &= p = 0.15 \\
 \sigma^2 &= \frac{p(1-p)}{n} \\
 &= \frac{0.15 \times 0.85}{200} \\
 &= 0.0006375 \\
 \sigma &= 0.02524876...
 \end{aligned}$$

$$\begin{aligned}
 \text{ii } z &= \frac{x - \mu}{\sigma} = \frac{0.2 - 0.15}{0.02525} \\
 &= 1.980198... = 1.98 \quad (2\text{dp}) \\
 P(z < 1.98) &= 0.9761 \\
 P(z > 1.98) &= 1 - 0.9761 \\
 &= 0.0239
 \end{aligned}$$

Q15

Sydney Tech

The heights in a population are normally distributed with a mean of 173 cm and a standard deviation of 7cm. Use the empirical rule to find the approximate probability that a randomly selected person has a height under 159 cm?



Q16

Penrith

A company manufactures lithium batteries using a mixture of digital and traditional techniques. Data shows the probability that a random case will fail quality control is 8%. An inspector selects a random batch of 50 cases from the warehouse.

Let X be the binomial random variable of the number of cases that do not pass the inspection.

- What is the mean, variance and standard deviation for this distribution, correct to 2 decimal places? 3
- Find the probability that the number of cases that fail to pass within one standard deviation of the mean. 2
- A new company standard insists that the number of failures in a batch must be no more than one standard deviation above the mean. Batches that fail to meet this standard are rejected. What is the probability of a batch being rejected? 2
- Due to the new regulations and the number of rejected batches, the company improves its manufacturing process so that the new experimental probability of failure is reduced to 4%. Find the new probability that a batch will be accepted. 2

$$\begin{aligned}
 X &\sim \text{Bin}(50, 0.08) \\
 \text{(i) } \mu &= E(X) = np \\
 &= 50 \times 0.08 \\
 &= 4 \\
 \text{Var}(X) &= npq \\
 &= 50 \times 0.08 \times 0.92 \\
 &= 3.68 \\
 \sigma &= \sqrt{3.68} = 1.92 \quad (2\text{d.p.}) \\
 \text{(ii) } P(\mu - \sigma \leq X \leq \mu + \sigma) \\
 &= P(2.08 \leq X \leq 5.92) \\
 &= P(X=3 \text{ or } X=4 \text{ or } X=5) \\
 &= {}^{50}C_3 (0.08)^3 (0.92)^{47} + {}^{50}C_4 (0.08)^4 (0.92)^{46} \\
 &\quad + {}^{50}C_5 (0.08)^5 (0.92)^{45} \\
 &= 0.5659 \quad (4\text{d.p.}) \quad \text{2 marks} \\
 \text{(iii) } P(X > 5) &= 1 - P(X=0 \text{ or } X=1 \text{ or } X=2 \text{ or } X=3 \text{ or } X=4 \text{ or } X=5) \\
 &= 1 - [{}^{50}C_0 (0.08)^0 (0.92)^{50} + {}^{50}C_1 (0.08)^1 (0.92)^{49} + {}^{50}C_2 (0.08)^2 (0.92)^{48} \\
 &\quad + {}^{50}C_3 (0.08)^3 (0.92)^{47} + {}^{50}C_4 (0.08)^4 (0.92)^{46} + {}^{50}C_5 (0.08)^5 (0.92)^{45}] \\
 &= 0.2081 \quad (4\text{d.p.}) \quad \text{2 marks} \\
 \text{(iv) } X &\sim \text{Bin}(50, 0.04) \\
 \mu &= 50 \times 0.04 = 2 \\
 \sigma &= \sqrt{50 \times 0.04 \times 0.96} = 1.39 \\
 \mu + \sigma &= 3.39 \\
 P(X \leq 3) \\
 &= {}^{50}C_0 (0.04)^0 (0.96)^{50} + {}^{50}C_1 (0.04)^1 (0.96)^{49} + {}^{50}C_2 (0.04)^2 (0.96)^{48} \\
 &\quad + {}^{50}C_3 (0.04)^3 (0.96)^{47} \\
 &= 0.8609 \quad (4\text{d.p.}) \quad \text{2 marks}
 \end{aligned}$$

2

7



Binomial Distribution

Part b: mean, variance and z score

3unit

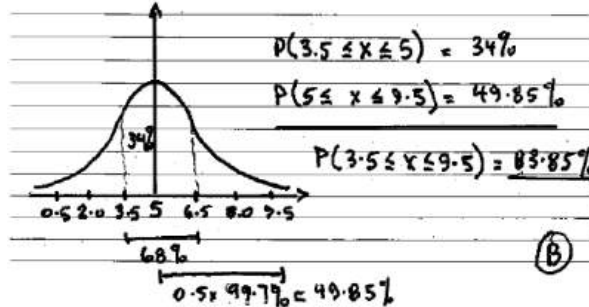
selective

Q17

Total Education

Calculate the probability for a normal distribution given $P(3.5 \leq X \leq 9.5)$ if $\mu = 5, \sigma = 1.5$.

- A. 67.65%
B. 83.85%
C. 84%
D. 99.7%



Q18

Baulkham Hills

The following table is the distribution of a random variable X where $X \sim \text{Bin}(4, p)$.

x	0	1	2	3	4
$P(X=x)$	0.2401	a	b	0.0756	0.0081

- (i) Use the known entries in the probability distribution table to calculate the value of p . 1
 (ii) Find the values of a and b . 2
 (iii) Calculate $\text{Var}(X)$. 1
 (iv) Find $P(X \geq 3)$. 1

(i) $P(X=4) = 0.0081$

$${}^4C_4 p^4 (1-p)^0 = 0.0081$$

$$p^4 = 0.0081$$

$$p = 0.3$$

(ii) $a = P(X=1)$

$$= {}^4C_1 \times 0.3^1 \times 0.7^3$$

$$= 0.4116$$

$$b = P(X=2)$$

$$= {}^4C_2 \times 0.3^2 \times 0.7^2$$

$$= 0.2646$$

Alternatively:

$$b = 1 - 0.2401 - 0.4116 - 0.0756 - 0.0081$$

$$= 0.2646$$

(iii) $\text{Var}(X) = npq$

$$= 4 \times 0.3 \times 0.7$$

$$= 0.84$$

(iv) $P(X \geq 3) = P(X=3) + P(X=4)$

$$= 0.0756 + 0.0081$$

$$= 0.0837$$

Q19

CSSA

What is the variance of the following Bernoulli distribution?

X	0	1
$P(X=x)$	0.2	0.8

- (A) 0.16
(B) 0.20
(C) 0.40
(D) 0.64

The variance of a Bernoulli distribution is given by:

$$\sigma^2 = p(1-p)$$

$$= 0.8(1-0.8)$$

$$= 0.16$$

Hence (A)

Q20

Enterprises

Find the variance for the Bernoulli distribution

$$p(x) = \begin{cases} 0.7 & \text{for } x=1 \\ 0.3 & \text{for } x=0 \end{cases} \quad \begin{matrix} P(1) = 0.7 \\ p = 0.7 \end{matrix}$$

$$\text{Var}(X) = p(1-p) = 0.7 \cdot 0.3 = 0.21$$

Q21

Total Education

Question 12 of the Extension 1 HSC Mathematics Examination has a maximum mark of 15. Approximately 7000 students attempt this question with the results being normally distributed with a mean $\mu=7.8$ marks and a standard deviation $\sigma=2.1$ marks.

- (i) What percentage of students would gain a mark of 12 or more? 2
 (ii) By showing a calculation, what word would you use to describe a mark of 15? 1

$$(i) Z = \frac{12 - 7.8}{2.1} = \frac{4.2}{2.1} = 2$$

$$P(X \geq 12) = \frac{5\%}{2} = 2.5\%$$

$$(ii) Z = \frac{15 - 7.8}{2.1} = \frac{7.2}{2.1} = 3.429 > 2.7$$

outlier



Binomial Distribution

Part C: Sample Proportion

3unit

selective

Q 1

Baulkham Hills

In a large population, the probability that a person is left-handed is 0.3. Let $\hat{p}$ be the random variable that represents the sample proportion of left-handed people from a sample size of n , drawn from the population.

What is the smallest value of n such that the standard deviation of $\hat{p}$ is less than or equal to 0.07?

- A. $n = 42$ B. $n = 43$ C. $n = 61$ D. $n = 62$

$$\sigma \leq 0.07$$

$$0.21 \leq n \times 0.07^2$$

$$\sqrt{\frac{pq}{n}} \leq 0.07$$

$$n \geq \frac{0.21}{0.07^2}$$

$$\sqrt{\frac{0.3 \times 0.7}{n}} \leq 0.07$$

The smallest value of n would be $n = 43$.

Q 2

Bonnyrigg

Mr. Tran is a professional basketball player who is known to score 70% of all his shots from the free throw line. He goes to the basketball court and takes 50 shots from the free throw line.

Use a normal approximation to the sample proportion to find the probability that Mr. Tran will score more than 85% of his shots (do not use continuity correction).

$$\hat{p} = 0.7 \text{ and } \sigma_{\hat{p}} = \sqrt{\frac{0.7 \times 0.3}{50}} \approx 0.065$$

$$P\left(\frac{X}{n} > 0.85\right) = P\left(Z > \frac{0.85 - 0.7}{0.065}\right) = P(Z > 2.31)$$

Using the table of z -scores:

$$P(Z > 2.31) = 1 - P(Z \leq 2.31) = 1 - 0.98956$$

$$\therefore P(Z > 2.31) \approx 0.01044 \text{ or } 1.04\%.$$

Q 3

Ace

It is known that 80% of patients with a certain disease can be cured with a certain drug.

What is the probability that amongst 150 patients with the disease, at most 37 of them cannot be cured with the drug? You must justify the use of the normal approximation.

$$n = 150, p = 0.8, q = 0.2$$

$$np = 120 > 10, nq = 30 > 10$$

probability can be approximated using the normal distribution.

$$\text{Solve } P(\hat{p} \geq \frac{113}{150}), \sigma = \sqrt{\frac{0.8 \times 0.2}{150}} = 0.033$$

$$E(\hat{p}) = 0.8, \therefore Z = \frac{\frac{113}{150} - 0.8}{0.033} = -1.43$$

$$\therefore P(Z \geq -1.43) = 0.9236 = 92.36\%$$

Q 4

Bonnyrigg

Fred owns a company that produces thousands of pegs each day. He randomly selects 41 pegs that are produced on one day and finds eight faulty pegs.

a. What is the proportion of faulty pegs in this sample?

b. Pegs are packed each day in boxes. Each box holds 12 pegs. Let $\hat{p}$ be the random variable that represents the proportion of faulty pegs in a box.

The actual proportion of faulty pegs produced by the company each day is $\frac{1}{6}$.

Find $\Pr\left(\hat{p} < \frac{1}{6}\right)$. Express your answer in the form $a(b)^n$, where a and b are positive rational numbers and n is a positive integer.

$$(a) \frac{8}{41}$$

$$(b) \hat{p} = \frac{8}{41}, E(X) = p = \frac{1}{6}, \sigma = \sqrt{\frac{(\frac{1}{6})(\frac{5}{6})}{12}} = 0.107$$

Since $np = 12 \times \frac{1}{6} = 2 < 5$ we cannot use a normal approximation.

$\therefore$ Let $X \sim B\left(12, \frac{1}{6}\right)$, where X := no. of faulty pegs.

$$\begin{aligned} P\left(\hat{p} < \frac{1}{6}\right) &= P(X < 2) = P(X = 0) + P(X = 1) \\ &= \left(\frac{5}{6}\right)^{12} + 12\left(\frac{5}{6}\right)^{11}\left(\frac{1}{6}\right)^1 \\ &= \left(\frac{17}{6}\right)\left(\frac{5}{6}\right)^{11} \end{aligned}$$

Q 5

CSSA

A standard quality check found that out of 200 light globes, 6 had a defect.

- If the sample proportion is approximately normally distributed, show that the mean and sample standard deviation are 0.03 and 0.01206 respectively. 2
- Find the z -score if 4 light globes in this sample are defective. 1
- Use the section of the table showing the values of $P(Z < z)$ to find the probability that the number of defective light globes in this sample was between 4 and 5. 2

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7258	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7518	0.7549
0.7	0.7580	0.7612	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7996	0.8023	0.8051	0.8079	0.8106	0.8133

$$(i) \hat{p} = \frac{6}{200} = 0.03$$

$$\mu_{\text{proportion}} = \hat{p} = 0.03$$

$$\sigma_{\text{proportion}} = \frac{\sigma}{n}$$

$$= \sqrt{\frac{np(1-p)}{n}}$$

$$= \sqrt{\frac{200 \times 0.03 \times 0.97}{200}}$$

$$= 0.01206 \text{ (5 decimal places)}$$

$$(ii) \text{ For 4 defective globes } \hat{p} = \frac{4}{200} = 0.02$$

$$z = \frac{0.02 - 0.03}{0.01206} = -0.83 \text{ (2 decimal places)}$$

(iii) To find $P(4 < X < 5)$ we need to calculate the z -score

$$x = \frac{5}{200} = 0.025$$

$$z = \frac{0.025 - 0.03}{0.01206} = -0.41 \text{ (2 decimal places)}$$

$$\therefore P(4 < X < 5) = P(-0.83 < z < -0.41)$$

$$= P(z < -0.41) - P(z < -0.83)$$

$$= (1 - P(z < 0.41)) - (1 - P(z < 0.83))$$

$$= (1 - 0.6591) - (1 - 0.7967)$$

$$= 0.1376$$



Binomial Distribution

Part C: Sample Proportion

3unit

selective

Q 6

Hurlstone

A state-wide housing study found that 36% of adults in NSW have a mortgage.

- (i) A random sample of 25 adults in NSW is to be taken to determine the proportion of those who have a mortgage.

Show that the mean and standard deviation for the distribution of sample proportions of such random samples are 0.36 and 0.096 respectively.

- (ii) Part of a table of $P(Z \leq z)$ values, where Z is a standard normal variable, is shown.

z	+0.00	+0.01	+0.02	+0.03	+0.04	+0.05	+0.06	+0.07	+0.08	+0.09
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964

Of a random sample of 25 adults in NSW, use the table to estimate the probability that at most three will have a mortgage.

Give your answer correct to four decimal places.

- (iii) If a random sample of 25 adults in NSW is taken, find the probability that the sample proportion is equal to the population proportion.

Give your answer correct to four decimal places.

$$(i) E(\hat{p}) = p = 0.36$$

$$sd(\hat{p}) = \sqrt{\frac{0.36 \times 0.64}{25}} = 0.096$$

ii) Transforming to a standard normal variable Z , gives:

$$P\left(Z < \frac{0.12 - 0.36}{0.096}\right) = P(Z < -2.5) = 1 - P(Z < 2.5) \\ = 1 - 0.9938 = 0.0062$$

iii)

The number of adults in the sample who have a mortgage is

$$25 \times 0.36 = 9.$$

Let X represent the number of adults who have a mortgage and

$$X \sim \text{Bin}(25, 0.36).$$

$$P(X = 9) = \binom{25}{9} (0.36)^9 (0.64)^{16} = 0.1644$$

Q 7

Killara

A restaurant knows from past experience that 27% of customers will order a dessert after the main course. For next week, the restaurant has taken 288 customer bookings.

Determine the probability that less than 100 will order dessert.

A normal distribution table is included overleaf.

Handwritten calculations for Q7:

$$p = 0.27$$

$$q = 0.73$$

$$\mu = np = 77.76$$

$$\sigma = \sqrt{np(1-p)} = \sqrt{56.7648} = 7.5342$$

$$z\text{-score} = \frac{99.5 - 77.76}{7.5342} = 2.95182$$

$$P(X < 100) \approx 0.998$$

Q 8

Sydney Tech

An unbiased coin is flipped 6400 times. The random variable X is the number of heads recorded.

- Assuming that X can be accurately approximated by a normal distribution, find the values of μ and σ such that $X \sim N(\mu, \sigma^2)$
- Find the z -scores of flipping 3260 heads and of flipping 3100 heads.
- Hence, use the table below to estimate the probability of flipping between 3100 and 3260 heads.

First Decimal Place										
z	.0	.1	.2	.3	.4	.5	.6	.7	.8	.9
0	0.5000	0.5398	0.5793	0.6179	0.6554	0.6915	0.7257	0.7580	0.7881	0.8159
1	0.8413	0.8643	0.8849	0.9032	0.9192	0.9332	0.9452	0.9554	0.9641	0.9713
2	0.9772	0.9821	0.9861	0.9893	0.9918	0.9938	0.9953	0.9965	0.9974	0.9981
3	0.9987	0.9990	0.9993	0.9995	0.9997	0.9998	0.9998	0.9999	0.9999	1.0000

Handwritten calculations for Q8:

$$i/ \mu = 6400 \times \frac{1}{2} = 3200$$

$$\sigma^2 = 6400 \times \frac{1}{2} \times \frac{1}{2} = 1600$$

$$\sigma = 40$$

ii/ For 3260: $z = \frac{3260 - 3200}{40} = 1.5$

For 3100: $z = \frac{3100 - 3200}{40} = -2.5$

iii/ $P(3100 \leq X \leq 3260)$

$$= P(-2.5 \leq Z \leq 1.5)$$

$$= P(Z \leq 1.5) - P(Z \leq -2.5)$$

$$= P(Z \leq 1.5) - (1 - P(Z \leq 2.5))$$

$$= 0.9332 - 1 + 0.9938 = 0.927$$

